Project Title:
Development of Cu-based catalyst for electrochemical CO_2 reduction to C_2 products: combined computational chemistry and electrochemical engineering approach

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March monthly report

1. Tasks list for March

- Investigation of eCO2RR performance under higher CO_2 flow rate (100-200ml/min) using sputtered Ag cathode
- Screening the eCO2RR performance of different single-atom catalysts (Co, Cu, Fe, Zn, and Ni-2D-NOC) under CO_2 gas stream of 150ml/min and j of 200mA/cm^2
- Synthesis of Ni-2D-NOC catalyst via pyrolysis method under Argon gas.
- Studies of influence of varied cathodic gas feed compositions on CO selectivity:
    a. Absolute CO_2 supply (15, 30, 100, 150 ml/min)
    b. Gas mixture of 15%CO_2/85%N_2 (total flow rate of 100ml/min)
    c. Gas mixture of 95%CO_2/5%O_2 (total flow rate of 100ml/min)

2. Experimental Procedure

Synthesis of Ni-2D-NOC catalyst via pyrolysis method
The synthesis of Ni-2D-NOC catalyst was completely followed the previous report.[1] Initially, Ni-Phc, Urea, and Dicy (weight ratio of **1:5:5**) were ground-mixed thoroughly together. Then, the mixture was placed in a quartz boat that was closed with a lid and pyrolyzed at 800 °C (ramp rate = 10 °C·min^{-1}) in a tube furnace under an Ar atmosphere (flow rate = 200 sccm) for 2h.

Preparation of Ni-2D-NOC cathode electrodes via spray coating method
The as-prepared Ni-2D-NOC catalyst was spray-coated onto Sigracet 36BB carbon paper. Briefly, 50 mg of the Ni-2D-NOC and 127µL or 10% by catalyst weight of XA-9 Binder were ultrasonically mixed with IPA for 30 min to form a dark colloidal solution which then was sprayed onto carbon paper with 0.5 ml/min on 95 °C hotplate to obtain ~ 0.5 ± 0.15 mg/cm^2.

Electrochemical CO2RR measurements
A 5 cm^2 MEA electrolyzer was used to evaluate the CO2RR performance under modulated operating conditions. The Ni foam, AEM (X37-50 Grade RT, Dioxide Materials), and sputtered Ag and M-2D-NOC (M denoted as Co, Cu, Fe, Zn, and Ni) cathode catalysts were employed as the anode, membrane, and cathode electrode, respectively. 50 ml of 0.5M KOH was recirculated and used throughout the experiment as the anolyte at 8 ml/min. Different humidified gas feed compositions was introduced into the cathodic inlet under various flow rates. The cathodic outlet was directly injected into an online gas-chromatography for further gas products analysis.

3. Results

Influence of CO_2 flow rate on eCO2RR performance of sputtered Ag cathode

In the previous report, we had conducted CO2RR at 5-50 mlCO_2/min and proved that the FE(CO) was CO_2 flow rate dependent with a maximum FE(CO) of 76% at 30 mlCO_2/min. To further extend the understanding of the influence of higher absolute CO_2 amounts, the CO2RR performance of the sputtered Ag was conducted at 100-200 mlCO_2/min. As shown in Fig. 1, we observed a slight raise in FE(CO) with approx. 5% of FE(H_2) at 100 mlCO_2/min comparing to 30 mlCO_2/min. By further elevating the CO_2 flow rate to 150 mlCO_2/min, the CO selectivity was drastically boosted in terms of achieving higher FE(CO) of approx. 89% with 5% FE(H_2), and reaching the equilibrium at 200 mlCO_2/min. These results demonstrated that the decreased kinetic of CO2RR at lower CO_2 supply might be attributed to the CO_2 deficiency on the catalytic active site of Ag which can be solved by increasing the CO_2 flow rate.[2]

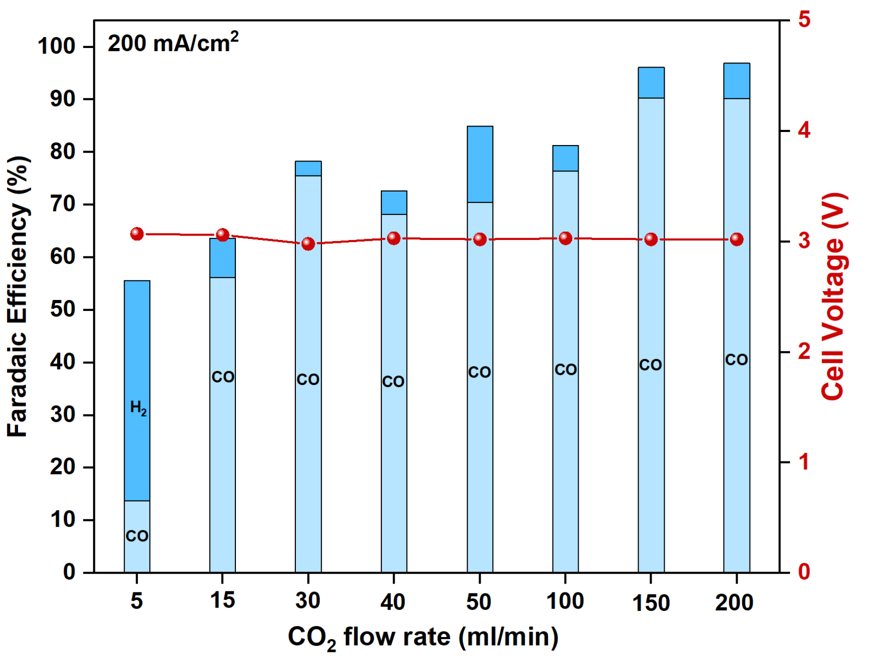

Figure 1. Faradaic efficiency of the MEA electrolyzer with sputtered Ag cathode under CO_2 flow rate range of 5-200 ml/min at 200mA/cm^2.

Influence of different cathodic gas feed compositions on CO selectivity of Ni-2D-NOC cathode catalyst

The CO2RR was performed at various absolute CO_2 flow rate (15, 30, 100, and 150 ml/min), j of 200mA/cm^2, and 0.5M KOH anolyte flow rate of 8 ml/min to identify the effect of CO_2 supply on overall MEA performance. A similar trend of poor CO selectivity to sputtered Ag was also noticed in Ni-2D-NOC cathode catalyst in which the current densities of both sputtered Ag and Ni-2D-NOC cathodes, whose FE(H_2) drastically increased at low-CO_2 supply, were strongly influenced by a competitive HER on the active sites due to the scarcity of surface-adsorbed CO_2. By further raising the amount of CO_2 from 15 to 30, 100, and 150 ml/min, the FE(CO) was exponentially increased with a maximum FE(CO) of approx. 98% at CO_2 supply of 150 ml/min, suggesting that the CO_2 to CO performance of Ni-2D-NOC catalyst was prone to correlating with the amount of CO_2. On the other hand, the direct conversion of low concentrated CO_2 feed is crucial for practical application with a direct flue gas that mainly contains of 5-40% CO_2 balanced by N_2 with trace amount of O_2 as the feedstock. (18 in low C). Diluted 15% CO_2/N_2 and 95%CO_2/5%O_2 gas mixtures were used as cathodic gas feed to determine the effect of low-concentrated CO_2 feed or low-P_{CO2} (CO_2 partial pressure) and competitiveness of oxygen reduction reaction (ORR) on CO selectivity of the MEA electrolyzer, respectively. As can be seen in Fig. 2, for a 15%CO_2/N_2 gas feed, the CO selectivity exhibited a similar trend with absolute CO_2 flow rate of 15 ml/min but even poorer CO2RR performance due to lower CO_2 concentration could lead to higher competing HER on the catalytic active sites (extrinsic factors, such as anolyte flow rate or water crossover management might play an essential role to suppress HER activity at low-P_{CO2}).[3–5] The oxygen reduction reaction (ORR) is a known catalytic reaction that can occur under reducing potentials and is thermodynamically more favorable than CO2RR to CO.[6] In the presence of 5%O_2 in 95%CO_2 cathodic gas feed, CO2RR performance and total faradaic efficiency were significantly detrimental by ORR activity. The CO selectivity was completely fell off after 1h of operating duration, suggested that high water content produced by ORR activity at the cathode surface could possibly enhance a competing HER activity and cause GDE flooding.

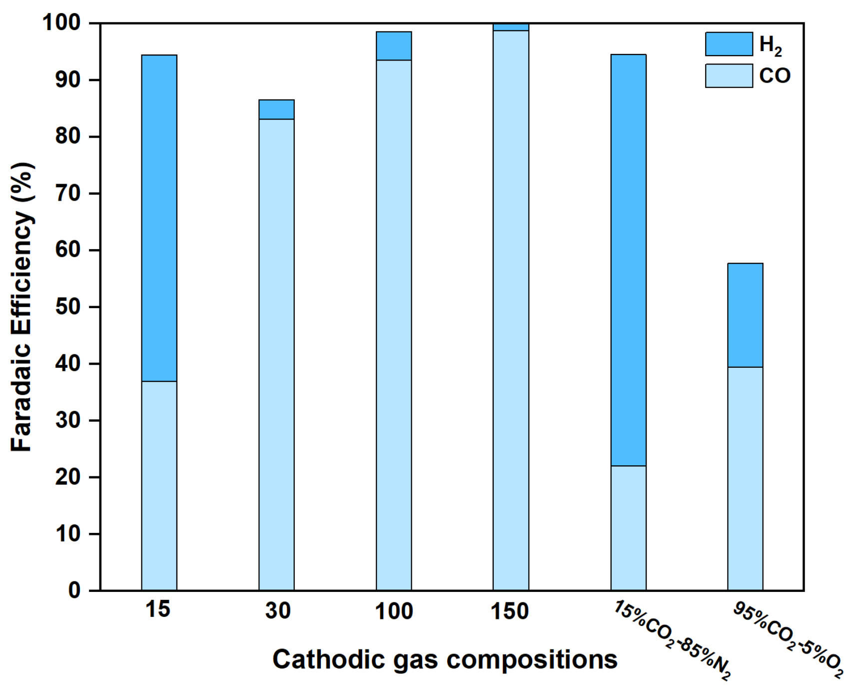

Figure 2. Faradaic Efficiency of MEA electrolyzer with Ni-2D-NOC cathode under 200mA/cm^2 and different cathodic gas compositions (15-150 mlCO_2/min, 15%CO_2/N_2, and 95%CO_2/O_2).

CO2RR performance of different cathode materials

The CO2RR performance of different Ag-based catalysts and M-2D-NOC single atom catalysts was prescreened under 200mA/cm^2 and 150mlCO_2/min. As shown in Fig. 3, Ag-based cathode catalysts exhibited a high CO selectivity in terms of achieving a maximum FE(CO) of approx. 89% with a slight H_2 byproduct. Meanwhile, most of as-prepared and exfoliated-M-2D NOC single atom catalysts (M = Co, Zn, Ni, Fe, and Cu), except the as-prepared Ni-2D-NOC, strongly suffered from the GDE flooding due to their high hydrophilicity, proton affinity, and catalytic HER which could lead to suppress CO2RR performance. On the other hands, the catalytic performance of Ni-2D-NOC seems unaffected by HER, showing superior CO_2 to CO activity compared to other materials, with a maximum FE(CO) of approx. 98%.

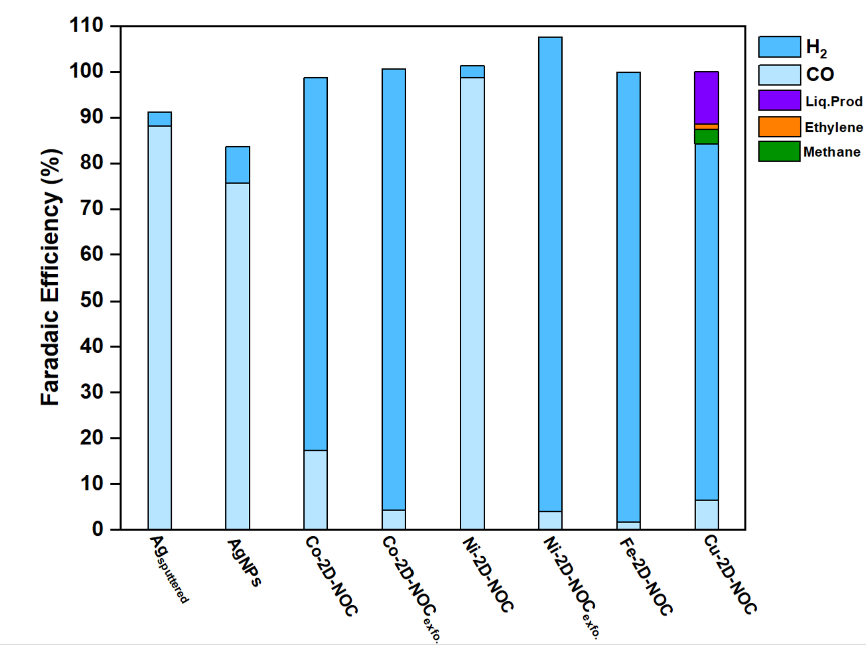

Figure 3. Faradaic efficiency of different cathode materials under applied current density of 200mA/cm^2 and 150 mlCO_2/min.

4. Summary

In summary, both sputtered Ag and Ni-2D-NOC cathode catalysts showed improved CO2RR performance when the absolute amount of CO_2 was increased. However, the Ni-2D-NOC exhibited poor CO_2 to CO activity at low absolute amounts of CO_2 and in a 15% CO_2/N_2 gas mixture due to a lack of surface-adsorbed CO_2 and competitive HER on the catalytic active sites. The addition of 5% O_2 in a 95% CO_2 gas feed caused severe CO2RR suppression but facilitated ORR activity due to its thermodynamic favorability. Evaluation of the CO2RR performance of M-2D-NOC single atom catalysts revealed that the as-prepared Ni-2D-NOC catalyst outperformed others in terms of high CO_2 to CO activity with a maximum FE(CO) of approx. 98%.

5. Upcoming tasks

- Synthesis of Ni_2-2D-NOC with a Ni-phc:Urea:Dicy ratio of 2:5:5
- CO2RR performance measurement of as-prepared Ni_2-2D-NOC
- Investigation of CO2RR performance of Ni-2D-NOC with different mass loading (0.65, 1, and 1.5 mg/cm^2)
- Literature review on Ni single atom catalysts for electrochemical reduction of CO_2 to CO

6. References

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